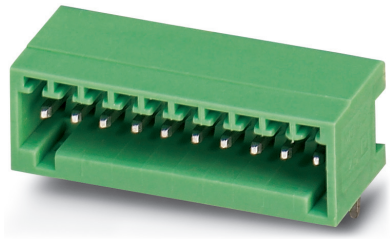
2021-02-26
Product version 00
Document revision 00

Data sheet

Order No.: 1881529

Type: MC 0,5/10-G-2,5

PCB header



1 Main features



- No. of pos. 10
- Nominal cross section 0.5 mm²
- Color green (6021)
- Pitch 2.5 mm
- Mounting type Wave soldering
- Nominal current 4 A
- Nominal voltage 160 V
- Connection direction 0 °
- Type of packaging packed in cardboard

2 Your advantages

- Well-known mounting principle allows worldwide use

Make sure you always use the latest documentation.
It can be downloaded at: phoenixcontact.net/product/1881529

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4 3D model in PDF can be activated (Acrobat Reader only)



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5 General Technical Data

5.1 item properties

Order No.	1881529
Type	MC 0,5/10-G-2,5
Plug-in system	MICRO COMBICON - FK-MC 0,5
Product type	PCB header
Type of contact	Male connector
Range of articles	MC 0,5/..-G
Pitch	2.5 mm
Number of positions	10
Number of levels	1
Number of connections	10
Number of potentials	10
Mounting type	Wave soldering
Connection direction of the connector to the PCB	0 °
Connection direction of the conductor to the PCB	0 °
Pin layout	Linear pinning
Solder pins per potential	1
Type	Standard

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6 Mounting

6.1 Flange mounting

Type of locking	without
Mounting flange	without

7 Material properties

7.1 Material of metal parts

Note	WEEE/RoHS-compliant, whisker-free acc. to IEC 60068-2-82/JEDEC JESD 201
Contact material	Cu alloy
Surface contact area	Nickel (1 - 3 µm Ni) , Tin (3 - 5 µm Sn)
Soldering area surface	Nickel (1 - 3 µm Ni) , Tin (3 - 5 µm Sn)
Surface characteristics	Tin-plated

Insulating material data	Housing
Color	green (6021)
Insulating material	PA
Insulating material group	I
CTI according to IEC 60112	600
Flammability rating according to UL 94	V0

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8 Dimensions

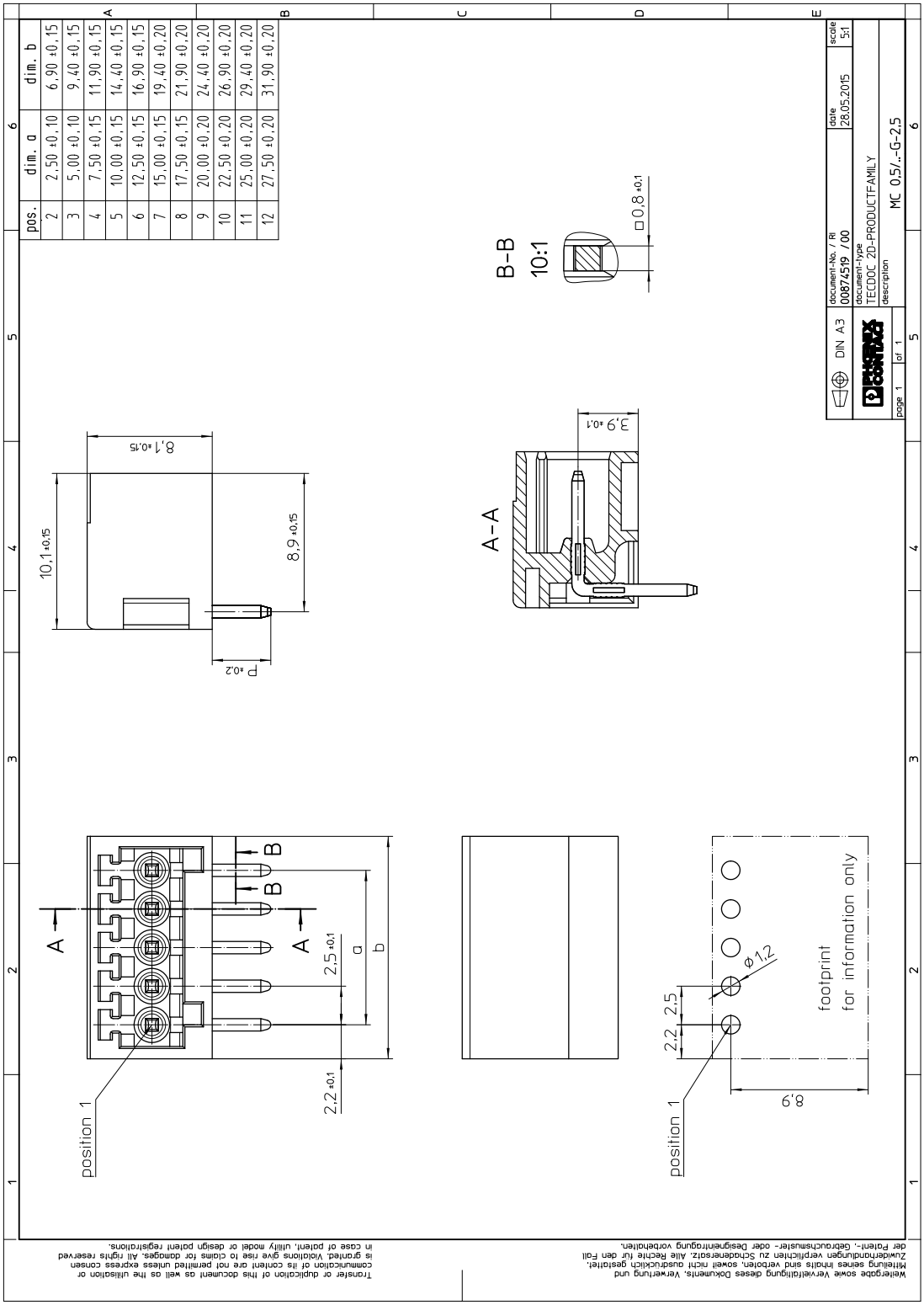
8.1 Dimensions for the product

Length	10.1 mm
Width	26.9 mm
Height (without solder pin)	8.1 mm
Total height	11.9 mm
Solder pin [P]	3.8 mm

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1881529 MC 0,5/10-G-2,5

9 Series drawing



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10 Application

11 Packaging information

Type of packaging	packed in cardboard
Pieces per package	50

11.1 Temperature limit values

Ambient temperature (storage/transport)	-40 °C ... 70 °C
Ambient temperature (assembly)	-5 °C ... 100 °C
Ambient temperature (operation)	-40 °C ... 100 °C (dependent on the derating curve)

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12 Mechanical tests

12.1 Visual examination

Specification	IEC 61984:2008-10
Visual examination	Test passed
Specification	IEC 60512-1-1:2002-02

12.2 Dimensional test

Dimensional test	Test passed
Specification	IEC 60512-1-2:2002-02

12.3 Resistance of marking

Resistance of marking	Test passed
Specification	IEC 60068-2-70:1995-12

12.4 Polarization and coding

Polarization and coding	Test passed
Specification	IEC 60512-13-5:2006-02
Test force	20 N

12.5 Contact retention in insert

Contact holder in insert Requirements >20 N	Test passed
Specification	IEC 60512-15-1:2008-05

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13 Insertion and withdrawal forces

Insertion and withdrawal force	Test passed
Specification	IEC 60512-13-2:2006-02
No. of cycles	25
Insertion strength per pos. approx.	8 N
Withdraw strength per pos. approx.	6 N

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14 Electrical tests

14.1 Electrical data

Rated current / conductor cross section	4 A / 0.5 mm²
Rated insulation voltage (III/2)	160 V
Rated surge voltage (III/2)	2.5 kV
Contact resistance	2 mΩ
Degree of pollution	2

14.2 Air and creepage distances

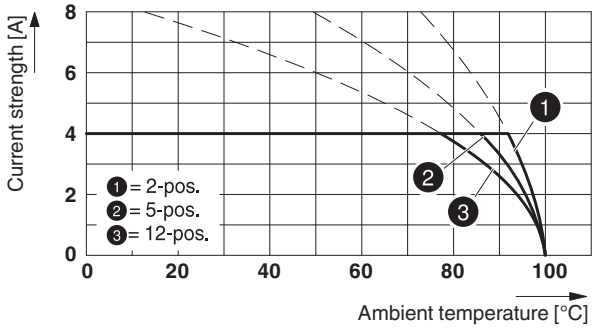
Component	PCB header		
Specification	IEC 60664-1:2007-04		
Mains type	unearthed mains		
Insulating material group	I		
Comparative tracking index (IEC 60112:2003-01)	CTI 600		
Rated insulation voltage	80 V	160 V	320 V
Rated surge voltage	1.5 kV	2.5 kV	2.5 kV
Degree of pollution	3	2	2
Overvoltage category	III	III	II
Minimum clearance case A (inhomogeneous field)	0.8 mm	1.5 mm	1.5 mm
Minimum value of the creepage path requirement in acc. with table	1.7 mm	0.8 mm	1.6 mm

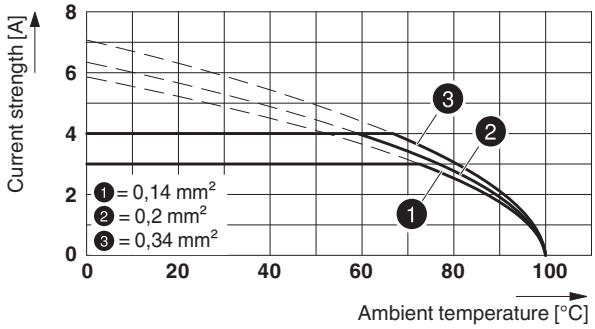
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15 Current carrying capacity/derating curves

Specification	IEC 61984:2008-10
Note	Representation based on IEC 60512-5-2:2002-02
Note	For number of positions, see diagram
Reduction factor	0.8
Conductor cross section	0.5 mm²

Type: FK-MC 0,5/...-ST-2,5 with MC 0,5/...-G-2,5





15.1 Vibration test

Specification	IEC 60068-2-6:2007-12
Result	Test passed
Frequency	10 - 150 - 10 Hz
Sweep speed	1 octave/min
Amplitude	0.35 mm (10 - 60.1 Hz)
Acceleration	5 g (60.1 - 150 Hz)
Test duration per axis	2.5 h
Test directions	X-, Y- and Z-axis
Note	The connected conductor loops were guided to the test sample at a distance of approx. 10 cm.

15.2 Insulation resistance

Specification	IEC 60512-3-1:2002-02
Result	Test passed
Insulation resistance, neighboring positions	> 5 MΩ

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16 Approvals / Certificates

IECEE CB Scheme	Voltage [V]	Current [A]	Cross section [AWG]	Cross section [mm²]
	80 V	4 A	-	-

EAC

VDE Gutachten mit Fertigungsüberwachung	Voltage [V]	Current [A]	Cross section [AWG]	Cross section [mm²]
	80 V	4 A	-	-

cULus Recognized	Voltage [V]	Current [A]	Cross section [AWG]	Cross section [mm²]
Usegroup B				
	125 V	4 A	-	-

CCA	Voltage [V]	Current [A]	Cross section [AWG]	Cross section [mm²]
	80 V	4 A	-	-

VDE Gutachten mit Fertigungsüberwachung	Voltage [V]	Current [A]	Cross section [AWG]	Cross section [mm²]
	80 V	4 A	-	-

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17 Commercial Data

Order No.	1881529
Type	MC 0,5/10-G-2,5
Pieces per package	50
Net weight	1.992 g
GTIN	4017918156763
	Information that applies locally, see link on page 1
Country of origin	Information that applies locally, see link on page 1

18 corresponding plugs

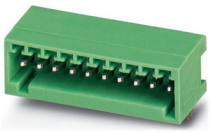
Order No.	Type
1881406	FK-MC 0,5/10-ST-2,5

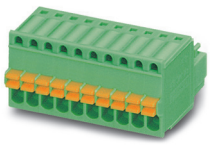
19 Accessories

Description	Order No.	Type
Coding profile, is inserted into the groove in the header, red insulating material	1881435	CP-MC 0,5
	0804853	SK 2,54/2,8:FORTL.ZAHLEN

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20 Combination tests





MC 0,5/..-G	**FK-MC 0,5/..-ST**
IEC 61984	IEC 61984
Mechanical tests (A)	
Insertion/withdrawal force per position	approx. 8 N / 6 N
Polarization when inserted Requirement >20 N	Test passed
Contact holder in insert Requirements >20 N	Test passed
Durability tests (B)	
Contact resistance R_1 1st level	2 mΩ
Contact resistance R_1 2nd level	
Insertion/withdrawal cycles	25
Contact resistance R_2	2.2 mΩ
Rated impulse voltage at sea level Voltage waveform ≥ (1.2/50 µs)	2.95 kV
Power-frequency withstand voltage Voltage waveform ≥ (50/60 Hz)	1.39 kV
Thermal tests (C)	
Tested number of positions	12
Tested conductor cross section	0.5 mm^2
Test current	4 A DC
Upper limiting temperature Requirements < 100°C	Test passed
Climatic tests (D)	
Test sequence 1: low temperature storage	-40 °C/2 h
Test sequence 2: heat storage	100 °C/168 h
Test sequence 3: noxious gas storage (ISO 6988)	0.2 dm^3 SO_2 on 300 dm^3/ 40 °C/1 cycle
Rated impulse voltage at sea level Voltage waveform ≥ (1.2/50 µs)	2.95 kV
Power-frequency withstand voltage Voltage waveform ≥ (50/60 Hz)	1.39 kV
Environmental and endurance tests (E)	
Specification	IEC 61984:2008-10
Degree of protection	Finger safety with IP20 test finger

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